

Problem

A 4,800-V rms transmission line feeds a distribution transformer with 1,200 turns on the primary and 28 turns on the secondary. When a 10-Ω load is connected across the secondary, find:

(a) the secondary voltage,

(b) the primary and secondary currents,

(c) the power supplied to the load.

Step-by-step solution

Step 1 of 3

$$(A) \quad \frac{V_2}{V_1} = \frac{N_2}{N_1} = n$$

$$V_2 = \left(\frac{N_2}{N_1} \right) V_1$$

$$= \left(\frac{28}{1200} \right) 4800$$

$$= 112 \text{ V} .$$

Step 2 of 3

$$(B) \quad I_2 = \frac{V_2}{R}$$

$$= \frac{112}{10} = 11.2 \text{ A}$$

And,

$$I_1 = n I_2 ,$$

$$n = \frac{28}{1200}$$

$$I_1 = \left(\frac{28}{1200} \right) 11.2 = 261.3 \text{ mA} .$$

Step 3 of 3

$$(C) \quad P = |I_2|^2 R = (11.2)^2 (10)$$

$$= 12.54 \text{ Watts} .$$